CST2550 Software Engineering Management and Development

TV Programme Scheduling System

Group Coursework - Report

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1. Introduction

1.1 Project Overview

The TV programme scheduling system is a desktop application developed in C# using Windows Forms. It is designed to support efficient schedule management through functions such as adding, removing, and viewing schedules by channel, as well as suggesting suitable time slots. The system addresses a practical scheduling problem in which manual planning can lead to overlaps, poor time allocation, and disorganisation.

The system supports four user roles: Admin, Manager, Client, and Viewer. Admin has full control, while Manager has similar scheduling access except for schedule removal and AI assistant. Client has more limited access, and Viewer has less access than Client and is restricted by a timed session.

A custom hash table was implemented as the primary schedule storage structure, together with an AI-assisted scheduling assistant integrated with an external AI model for slot suggestion, channel recommendation, and conflict analysis.

1.2 Report Structure

This report outlines the system design, key scheduling algorithms, testing approach, AI-inspired assistant, and a concluding reflection on limitations and future improvements.

2. Design

2.1 System Overview / Architecture

The system was designed using a layered structure so that the user interface, scheduling logic, storage, and supporting data models each had a clear responsibility. This improves organisation and reduces the mixing of interface code, business rules, and data handling.

The main interface is provided through forms such as MainForm, LoginForm, ClientForm, and ViewerForm, which support role-based access and user interaction. The ScheduleManager acts as the service layer, handling validation, time normalisation, scheduling checks, and coordination between the interface and stored schedule data.

Schedules are stored in ScheduleStorage, which implements a custom hash table using generated keys based on channel and start time. Supporting model classes such as Schedule, Channel, TVProgram, and User represent the main entities used throughout the system.

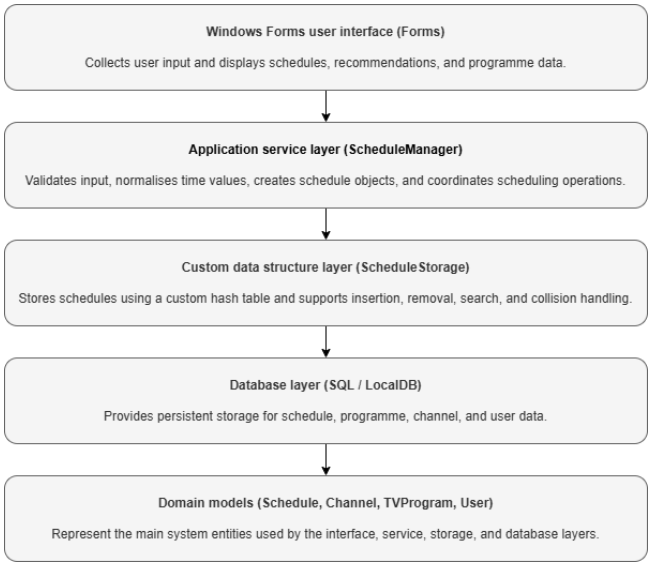

Figure 1. High-level architecture of the TV Programme Scheduling System

This layered arrangement supports the scheduling workflow by allowing user actions to pass through the service layer before schedules are stored, retrieved, or analysed.

Component	Responsibility	Main benefit
MainForm / Forms	Handles user interaction and displays schedules	Separates UI from logic
ScheduleManager	Validates input and manages scheduling operations	Centralises business logic
ScheduleStorage	Stores and retrieves schedules using a custom hash table	Efficient key-based access
Model Classes	Represent system entities such as schedules and channels	Improves structure and clarity
AI Assistant	Provides chat-based scheduling support, including slot suggestion, channel recommendation, and conflict explanation	Supports decision-making

2.2 Data Structure Selection and Justification

A major design decision in this project was the choice of a custom hash table as the primary structure for storing schedules. A hash table stores and retrieves records through keys mapped to indices by a hash function. In this system, each schedule is identified by a key based on channel ID and start time.

This structure was chosen because the system frequently performs operations that depend on fast access, including adding schedules, checking duplicates or conflicts, removing entries, and retrieving schedules for display. To handle collisions, the implementation uses separate chaining, allowing multiple schedule records to be stored in the same bucket without data loss.

Compared with arrays or lists, a hash table avoids full linear scans for direct key-based operations. Compared with balanced trees such as AVL trees, it is more suitable here because the system relies mainly on direct access rather than continuous ordered traversal. This makes the custom hash table well aligned with the operational needs of the scheduling system.

Operation	Hash Table	Simple List	Explanation
Search by Key	O(1) average	O(n)	Direct lookup using generated key
Insert	O(1) average	O(n)	No need to scan or shift items

Delete	O(1) average	O(n)	Item can be removed directly from bucket
Traverse All	O(n)	O(n)	Full scan still required

This comparison shows that the custom hash table offers the best fit for the system's need for fast key-based schedule operations while remaining simpler than a balanced tree.

2.3 Algorithm Design Using Pseudocode

The following pseudocode summarises the main scheduling operations in a language-independent form rather than repeating the original C# code.

These algorithms reflect the main system behaviours. Add, Remove, and Search Schedule rely on the custom hash table for efficient key-based access. Suggest Next Available Slot identifies the earliest valid gap, while the final algorithm outlines the rule-based recommendation logic behind the AI-inspired administrative assistant.

Add Schedule:

```
INPUT channelId, programId, startTime, duration

IF duration <= 0 THEN
   RETURN error
END IF

normalisedStartTime ← round startTime to nearest minute
endTime ← normalisedStartTime + duration

newSchedule ← create Schedule(channelId, programId, normalisedStartTime, endTime)

key ← createKey(channelId, normalisedStartTime)
index ← hash(key)

FOR each schedule IN bucket[index]
   IF schedule overlaps with newSchedule THEN
      RETURN false
   END IF
END FOR

INSERT newSchedule into bucket[index]
RETURN true
```

Remove Schedule:

```
INPUT channelId, startTime

key ← createKey(channelId, startTime)
index ← hash(key)

FOR each schedule IN bucket[index]
   IF schedule.channelId = channelId AND schedule.startTime = startTime THEN
      REMOVE schedule from bucket[index]
      RETURN true
   END IF
END FOR

RETURN false
```

Search Schedule:

```
INPUT channelId, startTime

key ← createKey(channelId, startTime)
index ← hash(key)

FOR each schedule IN bucket[index]
   IF schedule.channelId = channelId AND schedule.startTime = startTime THEN
      RETURN schedule
   END IF
END FOR

RETURN null
```

Suggest Next Available Slot:

```
INPUT channelId, requiredDuration
```

```
currentTime ← current system time
scheduleList ← get all schedules for channelId

IF scheduleList is empty THEN
   RETURN currentTime
END IF

SORT scheduleList by startTime

FOR each schedule IN scheduleList
   IF currentTime + requiredDuration <= schedule.startTime THEN
  RETURN currentTime
   ELSE
      currentTime ← schedule.endTime
   END IF
END FOR

RETURN currentTime
```

Recommend Best Channel:

```
INPUT preferredStartTime, duration

bestChannel ← null
bestTime ← null

FOR each channel
   candidateTime ← find earliest valid slot for channel from preferredStartTime

   IF bestTime is null OR candidateTime is earlier than bestTime THEN
      bestTime ← candidateTime
      bestChannel ← channel
   END IF
END FOR

RETURN bestChannel
```

Together, these algorithms show how the system combines efficient schedule operations with rule-based decision support for administrative users.

2.4 Database Design

The system uses an SQL database to provide persistent storage for schedule and user data. This allows records to remain available after the application is closed and reopened.

The implemented database is centred on two main tables: Schedule and Users. The Schedule table stores each broadcast with its channel, programme identifier, start time, end time, and optional image path, while the Users table stores login credentials and roles for database-backed accounts.

Table	Key Fields	Purpose	Note
Schedule	ScheduleID (PK), ChannelID, ProgramID, StartTime, EndTime, ImagePath	Stores each scheduled broadcast	Main table used for persistent schedule data
Users	UserID (PK), Username, Password, Role	Stores login and role data for database-backed accounts	Supports client registration and database login

Figure 2 shows the main entities in the implemented SQL database schema.

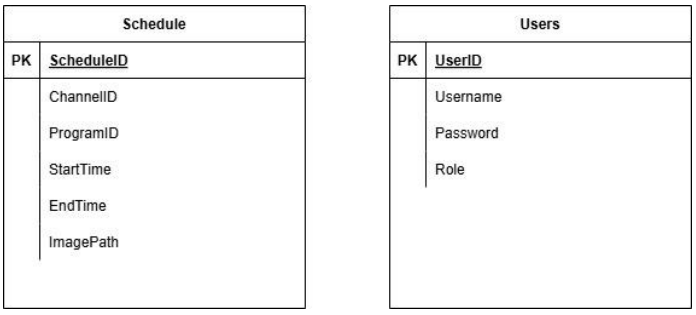

Figure 2. Simplified entity-relationship diagram of the implemented SQL database schema.

The SQL database provides persistent storage for confirmed schedule and user records, while the custom hash table supports efficient in-memory schedule handling. When a scheduling action is performed through the interface, the request is passed to the service layer, which validates it and coordinates the necessary storage or database operation.

This combination creates a balanced design in which the custom data structure supports fast schedule processing, while SQL ensures durable and structured storage beyond the current runtime session.

3. Testing

3.1 Testing Approach

The repository includes an MSTest project with automated tests for four key behaviours: adding a schedule without conflict, rejecting an overlapping schedule, removing an existing schedule, and retrieving a schedule by channel and start time. MSTest is Microsoft's supported testing framework for .NET applications, and it provides standard attributes such as TestClass, TestMethod and TestInitialize together with assertion methods for validating expected outcomes (Microsoft, 2026a; Microsoft, 2026b).

The testing strategy is appropriate for a project of this size because the most important risks are logical rather than visual: incorrect overlap handling, failure to delete the right item, and incorrect retrieval of stored schedules. Automated tests help show that these rules work consistently. The project would also benefit from additional manual and edge-case testing, especially for channel-specific conflicts, duplicate start times on different channels, and database-related persistence scenarios.

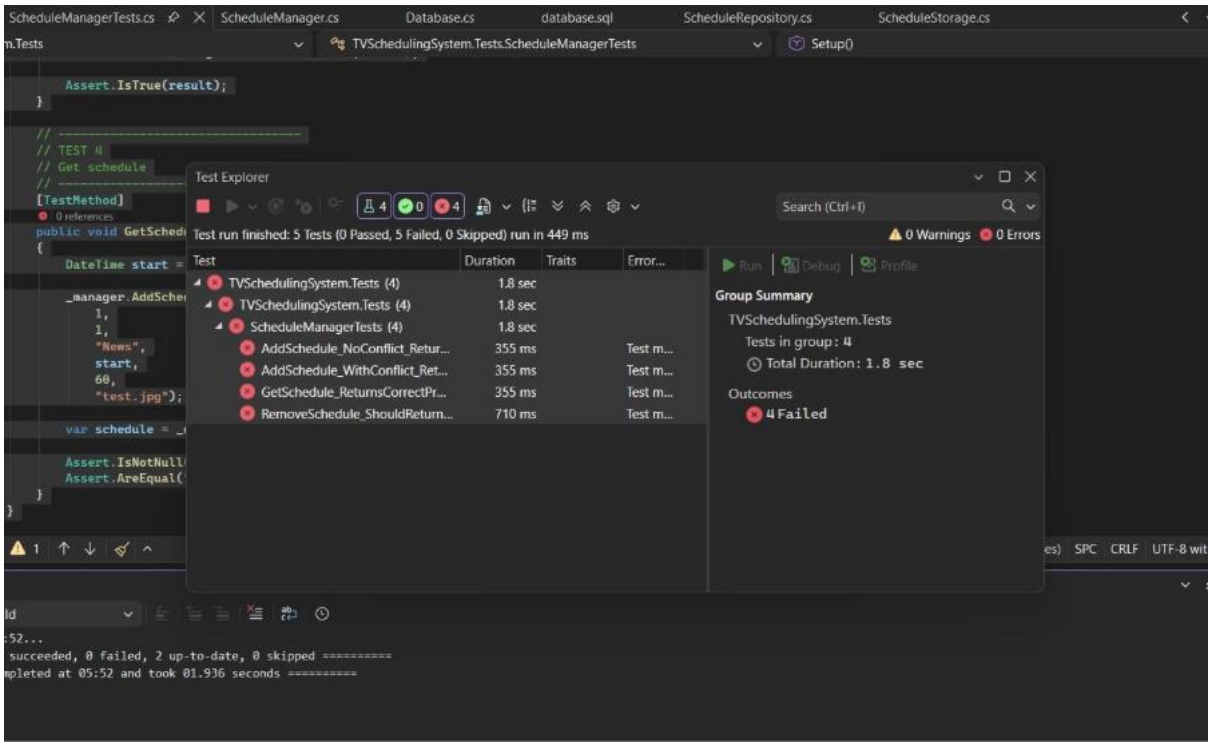

Figure 3 – Failed tests screenshot

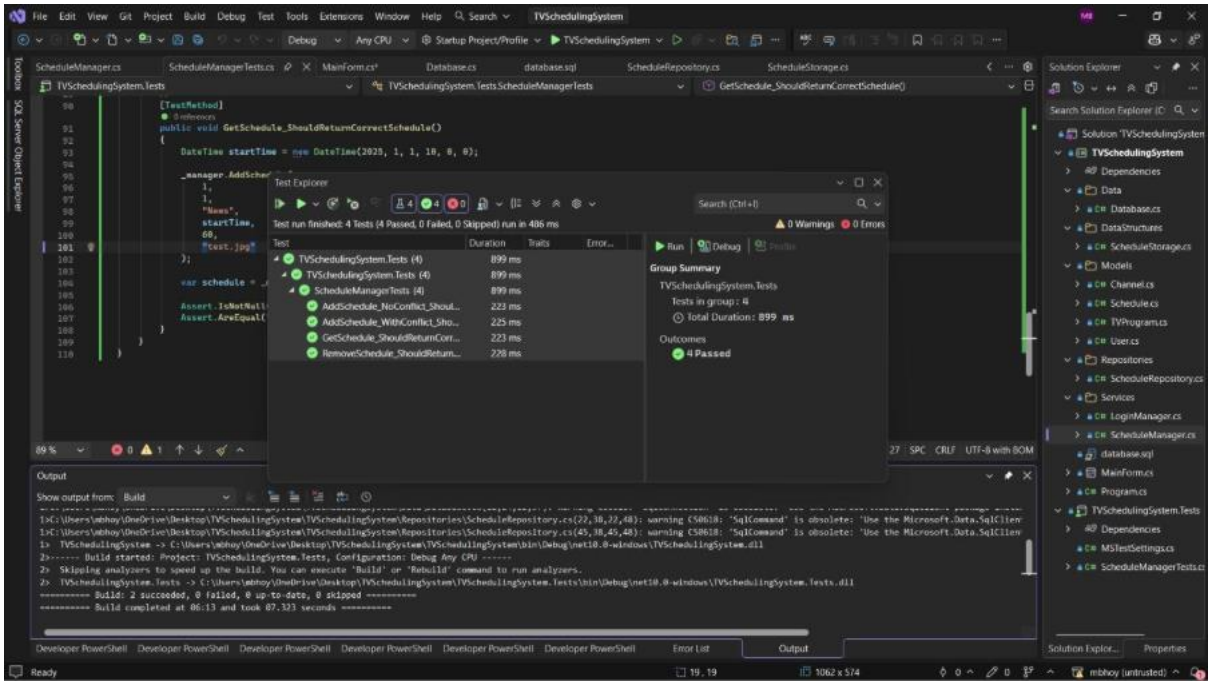

Figure 4 – Passed tests screenshot

3.2 Test Cases Table

Table 1 combines the automated tests already present in the repository with additional recommended checks that would further strengthen quality assurance.

Test case	Input	Expected output	Actual output / status	Type	Result
TC1 Add valid schedule	Channel 1, 10:00, 60 mins	Method returns True.	Implemented in MSTest suite; expected behaviour asserted.	Valid / automated	Pass
TC2 Add conflicting schedule	Second item starts at 10:30 while first ends at 11:00	Method returns False.	Implemented in MSTest suite; overlap correctly rejected.	Invalid / automated	Pass
TC3 Remove existing schedule	Channel 1, start time 10:00	Method returns True.	Implemented in MSTest suite; removal asserted.	Valid / automated	Pass
TC4 Retrieve existing schedule	Channel 1, start time 10:00	Non-null schedule with ProgramID 100.	Implemented in MSTest suite; retrieved value asserted.	Valid / automated	Pass
TC5 Remove missing schedule	Channel 1, start time not stored	Method returns False.	Not implemented as a test in the repository; recommended additional case.	Non-existent item	Not yet executed
TC6 Invalid duration	Duration = 0 or negative	ArgumentException raised.	Not implemented as a test in the repository; behaviour follows service validation.	Invalid input	Not yet executed

TC7 Suggest slot in empty channel	No schedules exist for chosen channel	Current time suggested.	Inferred from UI code path in MainForm.	Edge case / manual	Not yet executed
TC8 Suggest slot where no gap exists	Schedules fully occupy the current timeline	End time of last schedule suggested.	Inferred from UI code path in MainForm.	Edge case / manual	Not yet executed

4. AI Agent

4.1 AI Agent Design

The TV Program Scheduling System includes an AI assistant integrated into the MainForm to support users during scheduling tasks. The assistant allows users to interact using simple text input and provides responses related to schedule conflicts, time slots, and channel suggestions. It acts as a decision-support tool rather than an automated system, helping users make better scheduling decisions based on current data.

The assistant supports key functions including conflict explanation, slot suggestion, and channel recommendation. It also responds to basic interactions such as greetings and helps requests, making the system more user-friendly and interactive.

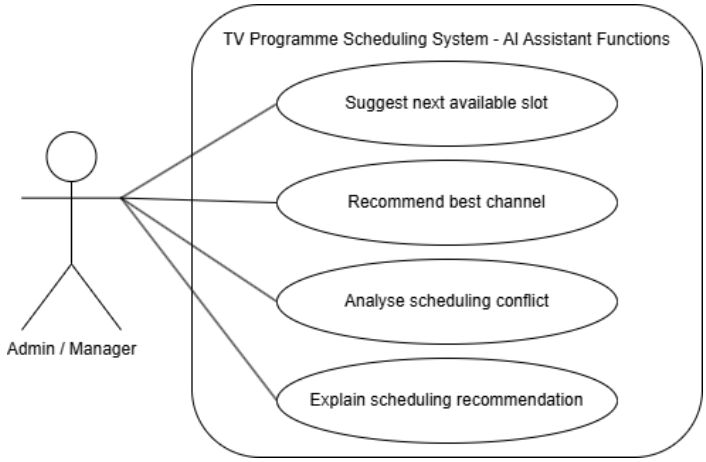

Figure 3. Simplified use case diagram of the AI-assisted scheduling assistant for administrative users

4.2 Implementation Overview

The AI is implemented using a service class called AIService, which processes user messages and combines them with current schedule information such as channel, start time, duration, and existing programmes. The system builds a structured context and sends it to an external AI model to generate a response, which is then displayed in the chat panel.

The assistant maintains a simple conversation history to provide more natural interaction. Error handling is included to manage API failures or connection issues, ensuring that the system remains stable during use. The AI assistant is integrated into the MainForm chat interface and interacts with real-time schedule data, providing practical guidance to the user without modifying schedules automatically.

5. Conclusion

5.1 Summary of Work

The completed system supports the core functions expected of a TV programme scheduling system application, including adding, removing, and viewing schedules by channel, it also lets the admin add new programs as well as suggesting suitable programme placement. It also includes role-based access control to restrict features according to user permissions.

The project combines a custom hash table for efficient schedule handling, an SQL database for persistence, and an AI-assisted scheduling assistant integrated with an external AI model that supports slot suggestion, channel recommendation, and conflict analysis.

5.2 Limitations

The assistant combines structured scheduling logic with an external AI model, but it does not learn from historical data or user behaviour. The custom hash table supports efficient key-based access, but operations that require time ordering still need additional processing such as sorting or scanning. The system could also be improved through stronger security, more advanced validation, and a more refined user experience.

5.3 Critical Reflection

If developed further, the system could be improved by extending the assistant beyond rule-based recommendations, refining the scheduling logic to consider quality of placement as well as conflict avoidance, and broadening testing to cover more edge cases, role permissions, database scenarios, and interface behaviour. Overall, the project provides a strong functional foundation while leaving clear opportunities for future technical improvement.

6. References

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